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What is CI/CD?

CI/CD stands for **Continuous Integration** and **Continuous Delivery/Continuous Deployment**. It is a modern software development practice that automates the entire process of building, testing, and releasing applications.

CI/CD serves as a core component of DevOps by bridging the gap between development and operations teams, enabling faster and more dependable software delivery.

Before CI/CD

In earlier software development practices, releasing applications was slow and heavily dependent on manual processes.

Developers typically worked on separate feature branches for long periods — sometimes weeks or even months — without integrating their changes frequently.

When all the branches were finally merged, it often led to major code conflicts and unstable builds. Since building and testing were done manually and usually at the final stage, bugs were discovered very late in the process. Fixing these late-stage issues was time-consuming, expensive, and stressful.

After CI/CD

With the introduction of CI/CD, software development became faster, more automated, and significantly more dependable. Developers now integrate their changes frequently into a shared main branch, reducing conflicts and ensuring smoother collaboration.

As soon as code is pushed, automated pipelines trigger builds, tests, and quality checks. This immediate feedback helps detect and fix issues early, preventing major problems later in the development cycle.

BEFORE CI/CD



WRITE CODE
(Separate Branches)



MANUAL BUILD



MANUAL TESTING



- Siloed Teams
- Slow Releases
- Risky Deployments

AFTER CI/CD



AUTOMATED BUILD



APPROVED
ARTIFACT



DEPLOY TO
PRODUCTION



- Frequent Releases
- Safer Deployments
- Immediate Feedback

Cloud Computing

When you use Google Drive, watch Netflix, send emails, or store photos — where is all that data stored?

It is not stored on your personal laptop.

It is stored on remote servers located in data centers around the world. That is the cloud.

Cloud computing means delivering computing services like servers, storage, databases, networking, and software over the internet instead of owning and managing physical infrastructure.

Importance of Cloud Computing

- **Scalability and Flexibility**

Cloud computing allows organizations to easily increase or decrease their computing resources based on demand. For example, if a website suddenly gets a lot of traffic, additional servers can be added instantly without buying new hardware.

- **Cost Efficiency**

Instead of investing large amounts of money in purchasing and maintaining physical infrastructure, businesses can use the cloud on a pay-as-you-go model. This reduces upfront costs and helps companies pay only for the resources they actually use.

- **High Availability and Reliability**

Cloud providers store data across multiple servers and data centers. This ensures that even if one server fails, the system continues running smoothly, improving reliability and reducing downtime.

- **Faster Deployment and Innovation**

Cloud services allow developers to quickly set up servers, databases, and applications within minutes. This speeds up development and helps organizations launch products and updates much faster.

- **Global Accessibility and Collaboration**

Cloud services can be accessed from anywhere through the internet. This allows teams to collaborate remotely, share data easily, and work efficiently from different locations.

On-prems vs clouds

On-Premises systems are hosted and managed within an organization's own data center, where the company is responsible for hardware, maintenance, and security. Cloud computing provides infrastructure and services over the internet through providers like Amazon Web Services or Microsoft Azure, reducing the need to manage physical infrastructure.

Issues with On-Prem that Cloud Solves

- **High Initial Cost** – On-prem requires heavy investment in hardware; cloud uses a pay-as-you-go model.
- **Limited Scalability** – Expanding on-prem requires new hardware; cloud resources scale instantly.
- **Maintenance Effort** – On-prem needs in-house management; cloud providers handle most maintenance.
- **Slow Deployment** – On-prem setup takes time; cloud services can be launched in minutes.

Service Models

Cloud services are generally divided into three main models:

- 1. IaaS (Infrastructure as a Service) You rent infrastructure like virtual machines and storage. You manage the OS and application.**
- 2. PaaS (Platform as a Service) You get a ready platform to build and deploy applications without worrying about infrastructure.**
- 3. SaaS (Software as a Service) You use software directly through the internet. Examples: Gmail, Zoom, Google Docs. give service models in more detail with slightly detailed definitions as well**